

Voice Emergency Checklist System

Generic advisory reader · audio-panel input · Config + Data cards

One combined reference: product, Aircraft Card spec, certification basis, hardware, and enclosure. Supersedes the separate tech, processor, and enclosure documents.

Cert path: NORSEE (PS-AIR-21.8-1602) · DO-160G environmental · deliberate DO-178C-avoidance argument · STC/PMA placement.

DEMO / TRAINING USE ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS

Not airworthy and not certified. The certification sections describe the path a productized unit would follow — a plan, not held approvals. No NORSEE LOA, STC, PMA, or TSOA exists for this article.

Scope & Status

DEMO / TRAINING USE ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS. This document describes a bench / training prototype and a certification-readiness roadmap. As built, the unit is not airworthy, not certified, and must never be installed in or relied upon aboard an aircraft. Nothing here is an STC, TSO authorization, PMA, or NORSEE letter of approval. The certification sections (Part C) describe the path a productized unit would follow — they are a plan, not a claim of compliance.

The product is a generic, offline, voice-driven advisory emergency-checklist reader. The pilot states an emergency by name; the device reads each checklist item aloud and waits for a spoken completion word before advancing. It runs entirely on-device (no cloud, no Wi-Fi). The product itself is aircraft-agnostic — all aircraft-specific behavior comes from the installed cards (Part B). The Cessna Citation CJ2 is included only as the reference example.

Crew audio is taken from the aircraft audio panel (analog or digital, selectable per installation), and recognition is voice-activated (VOX) by default — the pilot simply speaks, with a push-to-talk (PTT) button retained as a manual override. Configuration and checklist data live on two separate cards — a write-protected Config Card that defines the installation (aircraft selection, audio source, VOX behavior, hardware options) and a Data Card that carries the checklist library and audio. This two-card, audio-panel-fed design is a deliberate change from the earlier onboard-microphone / single-card demo and carries certification consequences addressed honestly in Part C.

This master reference supersedes and combines the previously separate documents: the technical data packet, the processor-selection note, and the enclosure specification. Those remain in the repository for history; this is the single source of truth.

Part	Contents
A Product	What the system is, the generic architecture, and the safety model
B The cards	The two-card (Config + Data) model + the formal card specifications & validation
C Certification basis	NORSEE / DO-160G / installation path, the deliberate DO-178C-avoidance argument, and the audio-panel-interface impact
D Hardware reference	Pin map, audio-input stage, VOX/PTT, per-device wiring, annunciator, lamp driver, power, BOM, processor selection
E Enclosure	Two-piece mechanical specification (dual card slots, isolated audio interface) for the fabricating engineer
F References	All cited regulatory and component sources

PART A

The Product

A generic, offline, advisory checklist reader. This part defines what the system is (and is not), its block-level architecture, and the three safety rules that the entire certification argument in Part C is built on.

A.1 What it is (and is not)

It is	It is not
An advisory read-aloud reader of checklist items	A required or primary aircraft system
Receive-only on the audio-panel INPUT tap (galvanically isolated, listen-only); plus a separate, galvanically-isolated OUTPUT into a dedicated COM3-style audio-panel channel for checklist read-aloud; no command/data to any aircraft system	A transmitter on aircraft COM radios, a panel control, or an interface to avionics/engines/flight controls. It does inject advisory audio into one dedicated aux/COM3-style channel via an isolated output — but that output is on a separate channel from the receive tap and is electrically isolated from required COM channels
Driven entirely by the installed Config Card + Data Card	Tied to one airframe in firmware
Offline, deterministic, single-chip	A cloud / connected / large-vocabulary STT device
A complement to the certified/required checklist	A substitute for the AFM/QRH or required checklist

This framing is not cosmetic — it is the foundation of the certification argument in Part C. The device connects to the aircraft through two electrically separate, galvanically-isolated channels: (1) a receive-only isolated INPUT tap (listen-only; cannot back-feed) used as the ASR speech source, and (2) a separate, galvanically-isolated OUTPUT into a dedicated COM3-style audio-panel channel (an isolated line-level transmit stage) that delivers checklist read-aloud audio to the crew in-headset. The input-tap language of “receive-only, isolated” is preserved for the INPUT path; the OUTPUT path is a deliberate, bounded, unidirectional injection into one dedicated aux channel. This dual-isolated-channel architecture is the interface posture the Part C argument is built on. Note that the addition of the isolated COM3-style audio output is the most significant certification and interface-risk change in this revision; Part C addresses it honestly.

A.2 Generic system architecture

Block	Function	Install-/aircraft-specific?
MCU + speech stack	VOX/wake word → command recognition → playback sequencing	No — fixed firmware
Audio-panel input stage	Takes crew speech from the aircraft audio panel — analog (isolated line tap) or digital (I2S codec), selectable per install	Config-driven — source set by the Config Card; path wired at install

Block	Function	Install-/aircraft-specific?
Audio-panel output stage (isolated TX to COM3-style channel)	Delivers checklist read-aloud audio out to a dedicated COM3-style audio-panel input channel via a galvanically-isolated, line-level output stage (isolation transformer + line driver on the TX line); crew hears checklist items in-headset. Onboard speaker/amplifier removed. The output is on a separate channel and separate connector from the receive-only INPUT tap.	No (fixed output stage hardware; the audio-panel COM3 channel it drives is chosen at installation)
Config Card (microSD, slot 1)	Defines the installation: active aircraft, audio source (analog/digital), VOX parameters, hardware options	Yes — per installation
Data Card (microSD, slot 2)	Carries the checklist library, trigger/advance vocabulary, and read-aloud audio	Yes — per aircraft
Annunciator switch	Dark-cockpit status / fault indication, IN/OUT select, PTT override	No

The MCU runs Espressif ESP-SR (AFE noise-suppression/VAD → WakeNet wake word → MultiNet fixed-grammar command recognition). The AFE's voice-activity detector (VAD) is what enables hands-free VOX: the pilot speaks and the device gates recognition on detected speech, with the PTT button retained as a manual override (force-listen). The grammar (trigger phrases, advance words) is small and bounded, which is what keeps an MCU-class part in scope (see D.7).

The crew-audio source is no longer an onboard microphone in the operational design — it is a tap off the aircraft audio panel. An onboard MEMS microphone is retained only as a documented bench-test option (D.3), never as the installed audio source.

A.3 The safety model (carried into the cert argument)

Three design rules define the failure behavior, and each one maps to a NORSEE requirement (Part C):

- 1 Revert-to-unopened. If the card is missing, unreadable, malformed, fails schema validation, or any referenced audio clip is absent, the device refuses to present any checklist and enters a clearly-annunciated FAULT state. It never shows partial or stale data. (*Failure mode = loss of function, not misleading information.*)
- 2 Dark-cockpit annunciation. When selected IN and healthy, the unit shows nothing. A fault lights the amber FAULT legend. Selected OUT shows the white OFF status and inhibits the fault legend (a deliberately deselected system needs no crew action). A power-up lamp test proves the legend is alive.

- 3 Dual-isolated-channel interface. The device's ties to the aircraft audio panel are two electrically separate, galvanically-isolated channels (C.3a, D.3): (a) a receive-only isolated INPUT tap — a high-impedance, isolation-transformer-coupled or buffered-receive-only tap off the audio panel; the unit cannot transmit, key, mute, or back-feed via this path; a short, open, or power loss inside the device cannot affect the panel via the input path; and (b) an isolated OUTPUT into a dedicated COM3-style channel — a galvanically-isolated, line-level output stage (isolation transformer on the TX line) that injects checklist read-aloud audio into the panel's dedicated COM3-style aux channel; physically separate connector from the input tap. The isolation barrier ensures a device fault cannot key, jam, load, or back-feed the panel's other channels or required COM radios. This output channel is the most significant certification and interface-risk item in this revision. The device draws power on its own protected rail. This dual-isolated-channel posture replaces the former "receive-only only" claim with a narrower, testable one covering both directions.

PART B

The Two Cards

Card defines everything — split across two. The product is generic; two installed microSD cards make it a specific aircraft's checklist reader, installed a specific way. A write-protected Config Card describes the installation; a Data Card carries the checklist content. This part is the formal specification — layouts, schemas, voice-grammar rules, and the validation logic that implements revert-to-unopened.

B.1 Principle

The product is generic. The two installed cards make it a specific aircraft's checklist reader, installed a specific way. Responsibility is split: the Config Card describes this installation (which aircraft, which audio source, how VOX behaves, what hardware is fitted); the Data Card carries the checklist content (library, vocabulary, audio). To support a new airframe you author a new Data Card; to re-use it in a different aircraft or wiring you change only the Config Card. No firmware change, no recompile.

Why two cards? Configuration is installation-controlled (set by the installer/shop and locked) while checklist data is content-controlled (authored from the AFM/QRH and revised as the source revises). Separating them keeps a content revision from silently changing the installation's audio/VOX setup, and lets a write-protected Config Card serve as the configuration-control record the FAA expects under NORSEE (Part C). Both cards are microSD/FAT32 and sit in two physical slots (slot 1 = CONFIG, slot 2 = DATA; see E.3).

B.2 Config Card (slot 1) — layout & schema

```
/sdcard-config/ (slot 1, microSD, FAT32)
config.json      the single installation-configuration manifest
```

config.json top-level object:

Field	Type	Req.	Meaning
schema_version	integer	rec.	Config-schema version the firmware validates against
aircraft	string	yes	Active aircraft id; must match a folder on the Data Card
audio_source	string	yes	analog or digital — selects the audio-panel input path (D.3.1)
audio	object	yes	Audio-input parameters (below)
vox	object	yes	VOX behavior (below)
hardware	object	opt.	Fitted-hardware options (codec part, legend rail, PTT present, etc.)
install	object	rec.	Provenance: shop, installer, date, work-order (configuration control)

audio object:

Field	Type	Req.	Meaning
input_gain_db	number	rec.	Input trim for the line-level tap (typ. 0–6 dB)

Field	Type	Req.	Meaning
codec	string	when digital	I2S codec fitted (e.g. es8388, es7210, pcm1808)
sample_rate_hz	integer	rec.	16000 for ESP-SR

vox object:

Field	Type	Req.	Meaning
mode	string	yes	vox (default, hands-free) or ptt_only (override-only)
vad_sensitivity	integer	rec.	AFE VAD aggressiveness 0–3 (higher = less false-trigger, may clip onset)
hangover_ms	integer	rec.	How long to keep listening after speech stops (debounce, typ. 300–600 ms)
ptt_override	boolean	rec.	true keeps PTT as a force-listen override even in vox mode

- ptt_only mode disables VOX and reverts to the legacy push-to-talk behavior.
- A Config Card whose aircraft has no matching Data Card folder is a fault (FAULT_AIRCRAFT_MISMATCH), not a silent guess.

B.3 Data Card (slot 2) — layout

```
/sdcard-data/  (slot 2, microSD, FAT32)
/              one folder per aircraft (e.g. CJ2, B200, PC12)
  checklists.json  the data manifest + checklist library
  audio/
    .wav           one read-aloud clip per referenced item + fault clips
```

- The firmware loads the folder named by the Config Card's aircraft field. If no Config Card is present at all, that is FAULT_NO_CONFIG — the device does not fall back to guessing a folder.

B.3a Data Card manifest schema (checklists.json)

Top-level object:

Field	Type	Req.	Meaning
schema_version	integer	rec.	Card schema version the firmware validates against
aircraft	string	yes	Short aircraft id; must match the folder name
title	string	yes	Human-readable aircraft name shown in logs
universal_advance	string[]	yes	Advance words accepted on every item (e.g. check, checked, complete, next)
checklists	object[]	yes	The checklist library (≥ 1)

Each checklists[] entry:

Field	Type	Req.	Meaning
id	string	yes	Stable identifier (e.g. engine_fire)
title	string	yes	Spoken/displayed checklist name
type	string	rec.	emergency, abnormal, normal
triggers	string[]	yes	Phrases that invoke this checklist by voice
items	object[]	yes	Ordered checklist steps (≥ 1)

Each items[] entry:

Field	Type	Req.	Meaning
clip	string	yes	Audio basename in audio/ (read-aloud <clip>.wav)
text	string	yes	The item text (for logs / optional display)
advance	string[]	opt.	Item-specific completion words (in addition to universal_advance)

B.3.1 Voice-grammar rules (validation-enforced)

MultiNet (English) imposes grammar constraints that the card author must honor; the validator rejects a card that violates them (revert-to-unopened):

- Phrases are lowercase letters and single spaces only — no digits or punctuation.
- Spell numbers as words: “V-one” → v one, not V1.
- Keep the total command count modest (well under the ~200-command MultiNet cap).
- Every clip referenced by any item must exist as audio/<clip>.wav.

B.4 Two-card validation & the revert-to-unopened rule

On boot the firmware attempts a complete, valid load of both cards and returns exactly one status. Both cards must be present, valid, and mutually consistent (the Config Card's aircraft must resolve to a Data Card folder):

Status	Card	Meaning	Annunc.
STORE_OK	both	Both cards mounted, both manifests parsed + schema-valid, every audio clip present, aircraft consistent	dark
FAULT_NO_CONFIG	config	Config Card not detected / config.json missing / unreadable	amber
FAULT_CONFIG_PARSE	config	config.json malformed	amber
FAULT_CONFIG_VALIDATION	config	Config schema / value violation (bad audio_source, bad vox.mode, missing required field)	amber
FAULT_NO_CARD	data	Data Card not detected / mount failed	amber
FAULT_AIRCRAFT_MISMATCH	both	Config aircraft has no matching Data Card folder	amber
FAULT_NO_JSON	data	checklists.json missing / unreadable	amber
FAULT_PARSE	data	Data JSON malformed	amber
FAULT_VALIDATION	data	Data schema or voice-rule violation, empty data	amber
FAULT_AUDIO_MISSING	data	A referenced clip is absent	amber
FAULT_NO_MEMORY	—	Allocation failed while loading	amber

On any FAULT the in-memory checklist table is left EMPTY — the device cannot present a partial or stale checklist even if asked. This is the literal implementation of “always revert to unopened if files are not available.” The boot order is Config Card first (it names the aircraft and the audio source the input stage must initialize), then the matching Data Card folder.

B.5 Card integrity & provenance (recommended for productized cards)

For cards that drive a *safety-enhancing* device, integrity matters as much as schema validity.

Recommended additions (forward-looking, beyond the current demo):

- Schema version gate — firmware refuses either card whose schema_version it does not support, rather than mis-parsing it.
- Manifest checksum / signature — a per-card hash (and, for production, a signature) so a corrupted or tampered card is rejected (FAULT_CONFIG_VALIDATION / FAULT_VALIDATION).
- Content provenance (Data Card) — record the AFM/QRH source revision the checklist text was transcribed from, the author, and the date.
- Installation provenance (Config Card) — the install object records shop, installer, date, and work-order. The Config Card is the per-tail configuration-control record the FAA expects under NORSEE (Part C): it documents *how this specific aircraft was set up*, including the audio source and VOX behavior.
- Read-only media — distribute production cards write-protected; the Config Card in particular should be locked after installation so configuration cannot drift in service.

PART C

Certification Basis

STC / PMA / NORSEE-ready reference. This part is a roadmap, not evidence of compliance — the demo unit holds no approvals. It explains why NORSEE is the right path, the standards elected as Minimum Design Requirements, and the deliberate, documented argument for avoiding DO-178C.

Status: this part is a roadmap, not evidence of compliance. The demo unit holds no approvals. The intent is to design and document so that a productized unit can pursue approval efficiently.

C.1 Why NORSEE is the right path

The product profile — non-required, advisory, independent of primary systems, failing to a clearly-annunciated safe state — matches the FAA's Non-Required Safety Enhancing Equipment (NORSEE) policy, PS-AIR-21.8-1602 (issued 03/31/2016)¹. NORSEE approval is a combined design and production approval issued under 14 CFR § 21.8(d) for equipment that is “not required by any Federal regulation with the intent to measurably increase aircraft safety,” determined to be a minor change to type design with a minor failure condition. A voice reader that simply reads aloud the same items already in the certified/required checklist — while the paper/electronic required checklist remains the authority — is a strong fit.

C.1.1 Applicability limit (read this)

NORSEE policy applies to aircraft certified under 14 CFR Part 23, 27, and 29 (and predecessor categories) and explicitly excludes Part 25 transport-category aircraft¹.

- The reference example, the Citation CJ2 (525A), is a Part 23 airplane — so NORSEE is available for it.
- Because the product is generic, the card library will inevitably include Part 25 aircraft (e.g. larger jets). For a Part 25 installation, NORSEE is not available; that installation must use an STC (or other major/minor-change path appropriate to the airframe). The product's certification status is therefore per-airframe, and the documentation must say so explicitly.

C.2 Minimum Design Requirements (the standards we elect)

Under NORSEE the applicant proposes the industry standard(s) that become the Minimum Design Requirements (MDR); the FAA accepts, partially accepts, or augments them. The FAA recommends widely-accepted RTCA / SAE / ASTM standards¹.

Topic	Elected standard / basis	Notes
Environmental	RTCA/DO-160G	Primary qualification evidence. Categories in C.4. FAA accepts DO-160 D/E/F/G per AC 21-16G and “strongly encourages DO-160G for new articles.”
Software	No DO-178C sought — see C.3	Argued out via function/failure classification, not avoided by omission
Complex hardware	DO-254 not invoked	COTS MCU + simple discrete logic; no custom complex devices (ASIC/FPGA/PLD). Per AC 20-152A, <i>simple</i> hardware verifiable by test needs no DO-254 design assurance.

Topic	Elected standard / basis	Notes
Human factors / color	FAA AC 25-11B conventions	amber=caution, white=status, dark-cockpit. Accepted design convention even though the unit is not a Part 25 article.
System safety (if needed)	SAE ARP4761 / AC 23.1309-1	Only invoked if the safety evaluation ever rises above "minor" (it should not)

C.3 The DO-178C–avoidance argument (deliberate, documented)

DO-178C is software *design-assurance* guidance whose rigor scales with the failure condition the software can contribute to. NORSEE lets us classify the function and, for a genuinely minor–failure advisory device, scope software assurance down to essentially nothing formal — provided the architecture earns it. The argument has four legs:

- 1 Function is advisory, not required. The device does not perform, command, or feed any required function. The crew's authority is the certified/required checklist; this unit only reads it aloud.
- 2 Worst-case failure is minor. The two credible failures are (a) loss of function — it goes silent or annunciates FAULT, and the crew uses the paper/QRH exactly as today; and (b) misleading information — mitigated by the revert-to-unopened rule (no partial/stale output), card validation, and a procedural requirement that the crew cross-checks against the required checklist. Neither failure reduces the crew's ability to cope with a condition worse than minor — the NORSEE safety-evaluation test¹.
- 3 Bounded dual-isolated interface. This revision adds two ties to an aircraft system (C.3a), both galvanically isolated: (a) a receive-only isolated INPUT tap off the audio panel — the device cannot command a function or produce output via this path; and (b) a dedicated isolated OUTPUT into a COM3-style audio-panel channel for checklist read-aloud delivery — a unidirectional audio injection through an isolation transformer, not a control/keying/command path. The device still produces no command or control output to any aircraft system, but it does inject advisory audio into one dedicated aux channel. The output injection is a more invasive interface than a pure receive-only tap and is called out as such; the argument now rests on *isolation + directionality + channel separation* rather than *total receive-only*¹.
- 4 Qualitative safety evaluation is permitted for non-complex equipment; a quantitative probabilistic analysis (and the DO-178C machinery that feeds it) is not required for a minor–failure advisory function¹.

Honest caveats (must stay in the doc): (1) This argument must be agreed with the FAA ACO early — the applicant proposes the classification; the FAA concurs. If the FAA judges the failure condition above minor (e.g. because of over-reliance / automation-dependency human factors, or because the audio-panel interface is judged to compromise an aircraft communication system), the program moves to §2 of the NORSEE policy (xx.1309, ARP4754A/ARP4761) and software assurance re-enters. (2) “No DO-178C” is earned by architecture and procedural mitigations, not by labeling — the revert-to-unopened behavior, the validation gate, the dark-cockpit annunciation, the dual-isolated-channel interface (receive-only input tap + isolated output into a dedicated COM3-style channel), and a mandatory limitation that the unit may not be used as a substitute for the required checklist are the price of that classification. (3) The audio-panel interface raises the installation bar — further still with the output channel. What was arguably a minor alteration (a self-contained box drawing only power) now wires into an aircraft communication system for both receive and transmit. The output channel is a more invasive interface than a pure receive-only tap, pushing the path toward an STC or careful field-approval scrutiny on most airframes (C.3a, C.5) — do not assume a logbook-entry minor alteration any more. (4) VOX adds a human-factors failure mode. Hands-free activation can false-trigger on ambient cockpit speech, ATC audio, or crew conversation, potentially reading an unrequested checklist; it is mitigated by VAD sensitivity tuning, a bounded trigger grammar, the retained PTT override, and the standing limitation that the required checklist remains the authority (C.3a, D.3.2).

C.3a Audio-panel interface impact (the honest part)

This revision adds two interfaces to the aircraft audio panel: a receive-only isolated INPUT tap (for crew-speech ASR) and a dedicated isolated OUTPUT channel (COM3-style) for checklist read-aloud

delivery. Both must be argued explicitly. The goal is to make each interface so narrow and so demonstrably bounded that the residual failure stays minor. The output path is the more significant of the two.

What the interface is — and is not:

Property	INPUT tap (receive-only)	OUTPUT channel (isolated TX to COM3)
Directionality	Receive-only. No path to transmit, key a radio, mute, or back-feed.	Output-only. Unidirectional line-level audio injection into the panel's dedicated COM3-style aux channel. Not a keying or control path.
Isolation	600 Ω aviation audio isolation transformer (e.g. Allen Avionics AGL series) on the analog path; buffered receive-only I2S for the digital path. No data driven back toward any aircraft bus.	Line-level isolation transformer on the TX output line; galvanic barrier between the device and the panel's COM3 input. A fault in the device cannot key, jam, load, or back-feed the panel's other channels or required COM radios.
Fault containment	Short, open, or power loss inside the device cannot load down, ground, or back-feed the panel via the input path.	Short, open, or power loss cannot key or jam the panel via the output path; the isolation transformer is the primary barrier.
Channel separation	Separate connector from the OUTPUT path; cannot be mis-mated.	Dedicated aux channel (COM3-style), not the crew intercom or any required COM radio channel. A failure on the output side cannot affect other panel channels.
No operational credit	The panel feed is <i>listened to</i> for recognition only.	The injected audio is advisory/informational only; the crew's authority is the required checklist.

Why this still supports a minor classification (INPUT tap):

- The audio panel and intercom continue to function identically whether the device is present, powered, or failed — the tap is parallel and high-impedance.
- The failure modes the input tap could plausibly add (loading, ground loop, injected noise) are removed by isolation and the receive-only topology, and are exactly what DO-160G conducted/induced-susceptibility and the audio-system installation tests are meant to verify (C.4).
- This is analogous to other listen-only cockpit aids (cockpit voice recorders, audio-logging headsets) that tap audio without compromising the source.

Why the output channel raises the bar (do not gloss over this):

- Isolation/failure modes. The isolation transformer on the TX line is the primary barrier; the isolation design ensures a fault inside the device (including output-stage failure, supply fault, or software runaway) cannot key, jam, load, or back-feed the panel's COM channels or intercom.
- Non-interference with required COM audio. The cert argument must show the injected advisory audio cannot interfere with required ATC/aircraft audio on the panel — e.g. that the output level is set conservatively, that the COM3 channel cannot bleed onto required COM1/COM2 channels, and that the injected audio cannot mask or be mistaken for required audio.
- Masking/intelligibility human factors. There must be no plausible confusion between the injected checklist audio and required ATC/aircraft audio, and the injected audio must not mask ATC calls. This requires

careful level treatment, a conservative output level, and the standing limitation that the required checklist remains the authority.

- The interface now wires into an aircraft communication system for both receive and transmit; plan for STC or carefully substantiated field approval that explicitly addresses the isolated output channel (C.5).

VOX (voice-activation) certification note. Replacing push-to-talk with VOX as the primary trigger introduces a false-activation human-factors mode (reading an unrequested checklist on stray speech / ATC audio). Mitigations carried into the design: bounded wake-word + trigger grammar (not open-vocabulary), tunable VAD sensitivity and hangover (Config Card, B.2), the retained PTT override, and the standing limitation that the required checklist remains the authority. The ACO will want this mode addressed in the safety / human-factors evaluation¹.

C.4 DO-160G environmental qualification plan

Categories a cockpit/avionics-bay unit would target. For the prototype these are design targets, not completed tests².

DO-160G section	Target category	Rationale for this unit
§4 Temperature & Altitude	Cat A2 (controlled/pressurized)	-15 °C to +55 °C operating; pressurized cabin
§5 Temperature Variation	Cat B	Cockpit rate of change
§6 Humidity	Cat A	Conformal coat recommended
§7 Operational/Crash Shock	Operational + crash-safety	Boards retained; nothing becomes a projectile
§8 Vibration	Cat S (fixed-wing)	Locking hardware; conformal coat; no press-fit-only parts
§15 Magnetic Effect	Class Z	Small device; classify by measured deflection
§16 Power Input	per installation (e.g. 28 VDC)	Only if a 28 V variant is built; TVS + fuse
§17 Voltage Spike	Cat A	Input transient protection
§18 AF Conducted Susceptibility	Cat —	As applicable to the supply; also relevant to the audio-panel input — verify recognition is not corrupted by AF conducted noise
§19 Induced Signal Susceptibility	Cat ZC	Cockpit; audio-input cabling routed/shielded per the panel's installation practice
§20 RF Susceptibility	Cat —	Aluminum case + grounded shield
§21 RF Emission	Cat M (or better)	Wi-Fi/BT disabled in firmware materially helps emissions
§22 Lightning Induced Transient	as installed	Behind-panel mounting reduces exposure
§25 ESD	per §25	Bond exposed metal; recessed connectors
§26 Flammability	UL94 V-0 materials	Required if not aluminum

The deliberate choice to disable Wi-Fi and Bluetooth in firmware is both a security decision and an emissions-qualification advantage (§21).

Audio-interface-specific evidence (beyond the table): because the unit now connects to the audio panel on both a receive-only INPUT tap and an isolated OUTPUT channel, qualification should additionally demonstrate — across all DO-160G conditions and including a failed/unpowered device — that (a) the receive tap does not degrade audio-panel performance (intercom level, sidetone, the panel's own VOX/hot-mic behavior), and (b) the isolated OUTPUT cannot key, jam, load, or back-feed the panel's other channels or required COM radios, and the injected advisory audio level does not mask or interfere with required ATC/aircraft audio. The isolation transformers on both paths (C.3a) and the high-impedance receive-only input topology are the design basis for that demonstration.

C.5 Approval & installation path (per airframe)

NORSEE approval is design + production approval — not installation approval. The full chain for a Part 23/27/29 airframe:

- 1 Pre-application — engage the responsible FAA ACO early; agree on the NORSEE classification (minor change / minor failure) and the MDR (C.2).
- 2 Design + test — complete DO-160G qualification, the safety evaluation, human-factors assessment, and the configuration-control / card-provenance package.
- 3 NORSEE Letter of Approval (LOA) under § 21.8(d), with the certifying statement of compliance (template in C.6) and a quality system per the policy.
- 4 Installation — NORSEE *eligibility* still requires an installation approval on each aircraft: if a minor alteration, a logbook entry / FAA Form 337 as applicable; if a major change to type design (panel structure, electrical integration), an STC or field-approval path.
- 5 Part 25 aircraft — skip NORSEE; pursue STC for the installation (C.1.1).

The audio-panel interface raises the install classification — the output channel raises it further. Because the device now wires into an aircraft communication system for both receive and transmit, step 4 should be approached assuming the interface makes the alteration more than minor on most airframes — i.e. plan for an STC or a field approval that specifically substantiates both the receive-only input tap and the isolated COM3-style audio output (isolated, no degradation of comm audio, no keying/jamming of required COM channels, no masking per C.3a/C.4), not a bare logbook entry. The earlier power-only/independent design could credibly claim a minor alteration; this one generally cannot.

C.5.1 Where TSO and PMA fit (and don't, here)

- TSO authorization is a *minimum-performance* design+production approval against a specific TSO. There is no TSO that defines a “voice advisory checklist reader,” so a TSOA is not the natural path; NORSEE (applicant-proposed MDR) is. A TSOA would also still require separate installation approval³.
- PMA is for *modification/replacement* articles for a type-certificated product and is typically tied to an STC or identity. This unit is added equipment, not a replacement part, so PMA is not the primary path — though a production unit may end up holding PMA in conjunction with an STC for specific airframes (14 CFR § 21.303).

C.6 Certifying statement of compliance (template)

Per the NORSEE policy, the LOA application carries a statement in this form (placeholders to be completed at application). *Reproduced as a template only — not a current certification.*

“I, (authorized representative), certify that (company) has complied with all applicable requirements as identified in RTCA/DO-160G and the other Minimum Design Requirements accepted for this article, and policy statement PS-AIR-21.8-1602, and that the article is produced under the required quality system.”

Two limitation statements the FAA expects on advisory NORSEE, to be placarded / in the manual:

"No operational credit may be taken for installation of this system." · "This system is not a required system and may not be used as a substitution for the certificated aircraft checklist."

C.7 Compliance summary matrix

Requirement	Means of compliance	Status (demo)
Non-required, safety-enhancing	NORSEE PS-AIR-21.8-1602, § 21.8(d)	Argued; not applied
Minor change / minor failure	Qualitative safety evaluation (C.3)	Drafted; ACO concurrence pending
Environmental	RTCA/DO-160G (C.4) per AC 21-16G	Targets defined; not tested
Software assurance	DO-178C not sought — advisory/minor (C.3)	Architecture supports it; ACO concurrence pending
Complex hardware	DO-254 not invoked (simple COTS) — AC 20-152A	N/A by design
Human factors / color	AC 25-11B conventions, dark-cockpit	Implemented in design
Audio-panel interface — INPUT tap	Receive-only + galvanic isolation (C.3a, C.4 §18/§19); cannot key/jam/back-feed via input path	Architecture defined; substantiation/test pending
Audio-panel interface — OUTPUT channel (COM3)	Galvanically-isolated line-level TX to dedicated COM3-style channel; isolation transformer on TX line; cannot key/jam/load panel or required COMs; advisory audio cannot mask required ATC audio (C.3a)	Architecture defined; this is the most significant new interface-risk item — isolation design + safety assessment required before productization
VOX false-activation	Bounded grammar + tunable VAD + retained PTT override (C.3a/D.3.2)	Mitigations defined; ACO human-factors concurrence pending
Installation	STC / substantiated field approval (audio-panel interface — both input tap and isolated COM3 output — is more than minor); minor-alteration unlikely	Per-aircraft; none performed
Part 25 airframes	STC (NORSEE excluded)	Flagged

¹ FAA Policy Statement PS-AIR-21.8-1602, "Approval of Non-Required Safety Enhancing Equipment (NORSEE)" — [gajsc.org](https://www.faa.gov/gajsc)

² RTCA/DO-160G; FAA AC 21-16G acceptance of DO-160 D-G — [faa.gov AC 21-16G](https://www.faa.gov/aircraft/air_cert)

³ FAA Technical Standard Orders (TSO) program — [faa.gov/aircraft/air_cert](https://www.faa.gov/aircraft/air_cert)

PART D

Hardware Reference

The generic hardware platform shared by every aircraft card: MCU, pin map, per-device wiring, annunciator, lamp driver, power budget, processor selection, firmware notes, and the DIY bill of materials. Pin numbers match firmware/main/board_pins.h.

Pin numbers match firmware/main/board_pins.h. The wiring diagram is in §D.10. Logic level is 3.3 V (the ESP32-S3 is not 5 V tolerant on GPIO).

D.1 System overview

Item	Value
MCU	ESP32-S3 (dual-core LX7 @ 240 MHz) — PSRAM required by ESP-SR
Recommended module	ESP32-S3-WROOM-1 N16R8 (16 MB flash, 8 MB octal PSRAM)
Speech stack	Espressif ESP-SR: AFE (NS/VAD → VOX) → WakeNet “Hi ESP” → MultiNet English
Crew audio input	From the aircraft audio panel on I2S_NUM_O, selectable per install: analog (isolated line tap → I2S codec ADC) or digital (I2S codec ADC fed from a buffered tap). Onboard MEMS mic = bench-test only
Activation	VOX (AFE VAD) primary, hands-free; PTT retained as manual override
Audio output	I2S_NUM_1 DAC → isolated line-level output stage (isolation transformer + line driver on the TX line) → dedicated COM3-style audio-panel input channel; crew hears checklist read-aloud in-headset. Onboard speaker/amplifier removed. A bench-test-only speaker output may optionally be provided in the development unit (not the installed configuration).
Config storage	two microSD slots — slot 1 Config Card, slot 2 Data Card (Part B), FAT32
Annunciation	Applied Avionics split-legend switch (dark-cockpit, AC 25-11B)

Two build paths: Integrated (ESP32-S3-Korvo-2 dev board — ES8311/ES7210 codec, microSD slot; line-in repurposed for the audio-panel receive feed; output stage wired to COM3 isolation transformer) or DIY (ESP32-S3 DevKitC-1 N16R8 + audio-panel input stage [isolation transformer + I2S codec ADC] + isolated audio output stage [I2S DAC → isolation transformer → line-level TX to COM3 channel] + two microSD breakouts + the annunciator switch). The earlier INMP441 MEMS mic and MAX98357A speaker-amp remain available only as bench-test items; neither is used in the installed configuration.

D.2 Master pin map

Function	Macro	GPIO	Dir	Notes
PTT override	PTT_GPIO	0	in (PU)	BOOT button; active-low; force-listen override of VOX
SELECT switch	SELECT_GPIO	10	in (PU)	IN = GPIO→GND (active-low)
Legend OFF (white)	LEGEND_OFF_GPIO	21	out	top legend half (via driver)

Function	Macro	GPIO	Dir	Notes
Legend FAULT (amber)	LEGEND_FAULT_GPIO	14	out	bottom legend half (via driver)
Status LED	STATUS_LED_GPIO	48	out	on-board RGB on most S3 devkits
Audio-in bit clock	AIN_BCLK_GPIO	4	out	I2SO BCLK → codec/mic SCK
Audio-in word select	AIN_LRCLK_GPIO	5	out	I2SO WS → codec/mic WS
Audio-in data	AIN_DIN_GPIO	6	in	codec ADC / mic SD → ESP DIN
Audio-in master clock	AIN_MCLK_GPIO	3	out	MCLK to the codec (ES8388/ES7210 need it; INMP441/PCM1808 do not)
Codec I2C SDA	CODEC_SDA_GPIO	1	i/o	ES-series codec control bus (digital path)
Codec I2C SCL	CODEC_SCL_GPIO	2	out	ES-series codec control bus (digital path)
SD clock	SD_CLK_GPIO	7	out	SDMMC CLK (shared by both card slots)
SD command	SD_CMD_GPIO	9	i/o	SDMMC CMD (needs pull-up)
SD data 0	SD_D0_GPIO	8	i/o	SDMMC DAT0 (needs pull-up)
Config-card detect	SD_CFG_CD_GPIO	47	in (PU)	slot-1 card-detect (Config Card)
Data-card detect	SD_DAT_CD_GPIO	38	in (PU)	slot-2 card-detect (Data Card)
Audio-out bit clock	AOUT_BCLK_GPIO	15	out	I2S1 BCLK → isolated audio-out (COM3) line driver/DAC
Audio-out word select	AOUT_LRCLK_GPIO	16	out	I2S1 WS → isolated audio-out (COM3) line driver/DAC
Audio-out data	AOUT_DOUT_GPIO	17	out	I2S1 DOUT → isolated audio-out (COM3) DIN; DAC → isolation transformer → line-level TX to COM3-style audio-panel channel. Bench-test note: a speaker-amp (e.g. MAX98357A) may be substituted here for bench testing only; not the installed output.

Polarity macros: PTT_ACTIVE_LOW=1, SELECT_ACTIVE_LOW=1, LEGEND_OFF_ACTIVE_HIGH=1, LEGEND_FAULT_ACTIVE_HIGH=1, STATUS_LED_ACTIVE_HIGH=1, LAMP_TEST_MS=2000. Set any unused output to -1 to disable it cleanly.

Two-slot SD note: the demo shares one SDMMC 1-bit bus (CLK/CMD/DAT0) across both card slots, distinguished by per-slot card-detect lines and by mounting each at its own path (/sdcard-config, /sdcard-data). A production unit may instead give each slot its own SPI or SDMMC bus to remove any contention; the firmware reads Config first, then Data (B.4).

Reserved / avoid pins (N16R8): GPIO33–37 (octal PSRAM/flash bus — do not use), GPIO19/20 (USB D–/D+), GPIO0/45/46 (strapping — must boot in the right state), GPIO26–32 (SPI flash on some modules). GPIO0 here is only the BOOT/PTT button, so it is safe. GPIO1/2/3 are used here for the codec I2C + MCLK; on the Korvo-2 use that board's BSP assignments instead.

D.3 Per-device wiring

D.3.1 Audio-panel input stage (the crew-audio source, I2S_NUM_0). Crew speech comes from the aircraft audio panel, not an onboard mic. The source is chosen by the Config Card (audio_source); both paths terminate as an I2S input to the ESP32-S3 and feed the ESP-SR AFE.

- Analog path (analog). Tap a headphone/intercom/line output from the panel (aviation audio is typically ~150–600 Ω , ~1–5 V RMS). Feed it through a 600 Ω audio ground-loop isolation transformer (e.g. Allen Avionics AGL series) for galvanic isolation, then through a ~220–470 Ω series resistor + simple RC anti-alias into the line-in of an I2S codec ADC (ES8388/ES7210 with MCLK on GPIO3, I2C control on GPIO1/2; or PCM1808/CS5343 which self-clock). The tap is high-impedance and parallel so the panel sees a negligible load; the device cannot back-feed the panel. Scale the divider so panel line level maps to the codec's full-scale input without clipping.
- Digital path (digital). Where the codec sits closer to the source, take a buffered, receive-only I2S/line feed into the same codec ADC. This is a receive-only I2S/line feed — BCLK/WS/MCLK are generated by the ESP32 for the ADC only, and no data line is driven back toward any aircraft bus *via this input path*. (The separate audio output to the COM3 channel is on I2S_NUM_1 via a dedicated isolated output stage; see D.3.3a.)
- Codec wiring: SCK→GPIO4, WS→GPIO5, ADC_DATA→GPIO6, MCLK→GPIO3, I2C SDA/SCL→GPIO1/2; supply per the codec (1.8–3.3 V analog/digital rails); decouple each rail 0.1 μ F.
- Bench-test option only: an INMP441 MEMS mic may be wired in place of the codec for desk testing (VDD→3V3, SCK→GPIO4, WS→GPIO5, SD→GPIO6, L/R→GND, 100 k Ω pulldown on SD, never clock with VDD off). This is not the installed audio source and must not be used in an aircraft (it does not hear the panel and breaks the receive-from-panel model).

D.3.2 VOX & PTT override. Recognition is gated by the AFE VAD (VOX): the device listens whenever speech is detected on the panel feed, tuned by the Config Card (vox.vad_sensitivity, vox.hangover_ms). The PTT button (GPIO0, active-low) is a force-listen override — holding it opens recognition regardless of VAD, and with vox.mode = ptt_only it becomes the sole trigger (VOX disabled). PTT is debounced in software. There is no PTT/keying line toward the aircraft — this button only tells the device's own recognizer to listen.

D.3.3a Isolated audio output stage → COM3-style audio-panel channel (I2S_NUM_1). The installed audio output is a galvanically-isolated, line-level output stage on I2S_NUM_1 (GPIO15 BCLK, GPIO16 LRC, GPIO17 DOUT). The output chain is: ESP32-S3 I2S DAC → I2S-to-analog DAC/line driver → a line-level isolation transformer on the TX output line → line-level output (600 Ω nominal or per the panel's COM3 input impedance) → dedicated COM3-style audio-panel input channel. The isolation transformer on the TX line provides galvanic isolation; a device fault (short, open, supply failure, output-stage failure) cannot key, jam, load, or back-feed the panel's other channels or required COM radios. Output level should be set conservatively so the injected advisory audio is clearly audible but does not mask required ATC/aircraft audio. *Judgment call / open item*: the exact I2S DAC part and isolation transformer type for the output stage must be confirmed by the bench engineer. Options include a small I2S DAC IC (e.g. PCM5102A class) followed by a 600 Ω :600 Ω aviation audio isolation transformer (Allen Avionics AGL series or equivalent). The AOUT_BCLK/LRCLK/DOUT macros (GPIO15/16/17) match the former speaker-amp I2S assignments; the firmware I2S_NUM_1 driver is retained — only the physical output hardware changes.

D.3.3b Bench-test speaker output (development/test only — not the installed output). For bench verification of audio content before the isolated output stage is fitted, a MAX98357A I2S Class-D amp may be wired on GPIO15/16/17: supply 2.5–5.5 V; ~2.4 mA quiescent; peak ~650 mA at 5 V/4 Ω ; no MCLK. VIN→5 V, GND→GND,

BCLK→GPIO15, LRC→GPIO16, DIN→GPIO17, GAIN NC = 9 dB, SD/mode float = mono. OUT+/OUT- are bridge-tied — never to GND. This bench speaker is not routed to the aircraft audio panel and is NOT the installed audio output.

D.3.4 Two microSD card slots → SDMMC 1-bit. 3.3 V cards. Shared bus CLK→GPIO7, CMD→GPIO9 (10 kΩ→3V3), DAT0→GPIO8 (10 kΩ→3V3), VDD→3V3, VSS→GND. Slot 1 = Config Card (detect on GPIO47, mount /sdcard-config), slot 2 = Data Card (detect on GPIO38, mount /sdcard-data). FAT32; layouts per Part B. (Production option: separate bus per slot.)

D.3.5 Discrete inputs. SELECT (GPIO10): LOW = selected IN; pull-up HIGH = OUT. PTT (GPIO0): LOW = pressed (override). Each is a simple SPST to GND; debounce in software.

D.4 Annunciator switch (split-legend, dark-cockpit)

State	TOP — white VOICE CHKLST OFF	BOTTOM — amber VOICE CHKLST FAULT
Selected IN, healthy	dark	dark ← true dark cockpit
Selected IN, fault	dark	amber ON
Selected OUT	white ON	dark (fault inhibited)

Power-up lamp test: both halves on for ~2 s, then dark, so a dead LED cannot masquerade as healthy. OUT inhibits fault per AC 25-11B — a deselected system needs no crew action, so only the white OFF status shows; recognition and fault monitoring are suspended while OUT.

D.5 Lamp-driver circuit (one per legend half)

An ESP32 GPIO (~20 mA default, 40 mA max; 1.5 A total chip limit) cannot drive a 28 V — or 5 V at lamp current — legend directly. Use a low-side switch per half:

```

+V_lamp (5 V or 28 V, separate rail)
|
[ legend lamp ]      <- VIVISUN legend (white or amber)
|
+----- Drain
GPIO21 --[1k]--|G   N-ch MOSFET (logic-level, e.g. 2N7002 / A03400)
(or G14)      |   or NPN (e.g. 2N2222 with base resistor)
[10k]         <- gate/base pulldown -> GND (defined OFF)
|
GND (Source) ----- common ground with ESP32

```

One driver for GPIO21 (OFF/white), one for GPIO14 (FAULT/amber). The 10 kΩ pulldown guarantees the lamp is OFF during boot/reset before the GPIO is configured (dark-cockpit integrity). For a 28 V incandescent legend, use a FET with $V_{ds} \geq 40$ V and current $\geq$ inrush. Applied Avionics VIVISUN/Korry legends come in 28 VDC, 5 VDC, 28 VAC, 5 VAC, and 115 V lamp variants. For a quick bench mock-up, substitute two 3.3 V LEDs (white + amber) with series resistors driven straight from GPIO21/GPIO14 within the ~20 mA limit.

D.6 Power budget

Rail	Loads	Typical	Peak
3.3 V	ESP32-S3 (Wi-Fi off) + audio codec + audio output DAC/line driver + two microSD slots	~90–180 mA	~290 mA (SD init / SR burst)
5 V	Audio output line driver (if used; typically low-current line-level stage)	a few mA	~50 mA (varies by line driver; not the ~650 mA speaker-amp figure — speaker amp removed)
Lamp rail	up to 2 legend halves	0 (dark)	per lamp spec (e.g. 28 V incand.)

The audio-input codec (a few mA–~20 mA) and the second microSD slot add a little to the 3.3 V rail; the isolation transformers (input and output) are passive. Power from USB 5 V $\geq$ 1 A (the removed MAX98357A was the dominant load; the new output stage is much lower current). Keep a 28 V legend supply separate from logic 5 V (grounds common only). Bulk decoupling $\geq$ 100 μ F near the output line driver (if applicable) plus 0.1 μ F per device. The audio-panel tap draws no power from the aircraft and is isolated from the device's own grounds through the transformer (analog path); the audio-panel output stage is similarly isolated.

D.7 Processor selection

This is a narrow, bounded workload — a small fixed vocabulary (checklist triggers + universal advance words, well under the ~200-command MultiNet cap), offline, and deterministic. That is command recognition on a fixed grammar, not open-ended transcription, which keeps an MCU-class part firmly in scope and makes a Linux SBC overkill. It is also exactly the property that supports the minor-failure / no-DO-178C argument (C.3): a single-chip, fixed-grammar device has a small, well-understood failure surface.

Recommendation: keep the ESP32-S3 as the baseline; consider the ESP32-P4 only for headroom. ESP-SR (v2.1+) supports S3 and P4 for English MultiNet; the classic ESP32 is no longer supported by the current speech algorithms and should be avoided.

Option	Summary	Verdict
ESP32-S3 (baseline)	LX7 dual-core @240 MHz + AI vector ext, 512 KB SRAM, Wi-Fi+BLE, ~\$8–15	Best AI/voice part in the line, most mature tooling; firmware/diagram/BOM built on it
ESP32-P4 (upgrade)	RISC-V to 400 MHz + AI ext, 768 KB SRAM, ~2.5 $\times$ compute, no Wi-Fi/BT	Worth it only for more compute or a future display; no-radio is arguably a safety plus
Syntiant NDP120	Always-on ultra-low-power NDP	Overkill; we have panel power, need full grammar + playback + SD
Picovoice Porcupine+Rhino	Offline wake+intent on Cortex-M4	Clean fit but needs a per-deployment license key — undesirable for self-contained safety gear

Option	Summary	Verdict
Fluent.ai / NXP i.MX RT600	Commercial intent / heavy audio DSP	More licensing/board complexity than a fixed grammar needs
Raspberry Pi (whisper/Vosk)	Full Linux STT	Non-deterministic boot, higher power/cost — less robust for a fixed-grammar safety device

D.8 Firmware build notes

Topic	Value
Framework	ESP-IDF ≥ 5.2
Components	esp-sr, esp_spiffs, driver, json (cJSON), fatfs, sdmmc, esp_driver_sdmmc, esp_codec_dev (codec init for the digital/analog path)
Speech models	WakeNet WN9_HIESP; MultiNet English mn6_en/mn7_en (S3 only)
Partitions	factory app 3 MB + model 5 MB + storage 2 MB → needs 16 MB flash (N16R8)
Audio input	ESP-SR AFE fed from I2SO (codec ADC); VAD→VOX gating, PTT override; codec init from Config Card audio.codec
Two-card load	Config Card first (/sdcard-config/config.json) → init audio source + VOX → then matching Data Card folder (/sdcard-data/<aircraft>/)
Grammar rules	lowercase + single spaces; spell numbers (“v one”); ~200-cmd cap
Fault behavior	any card/config fault → ST_FAULT, amber legend, no checklist shown (B.4)

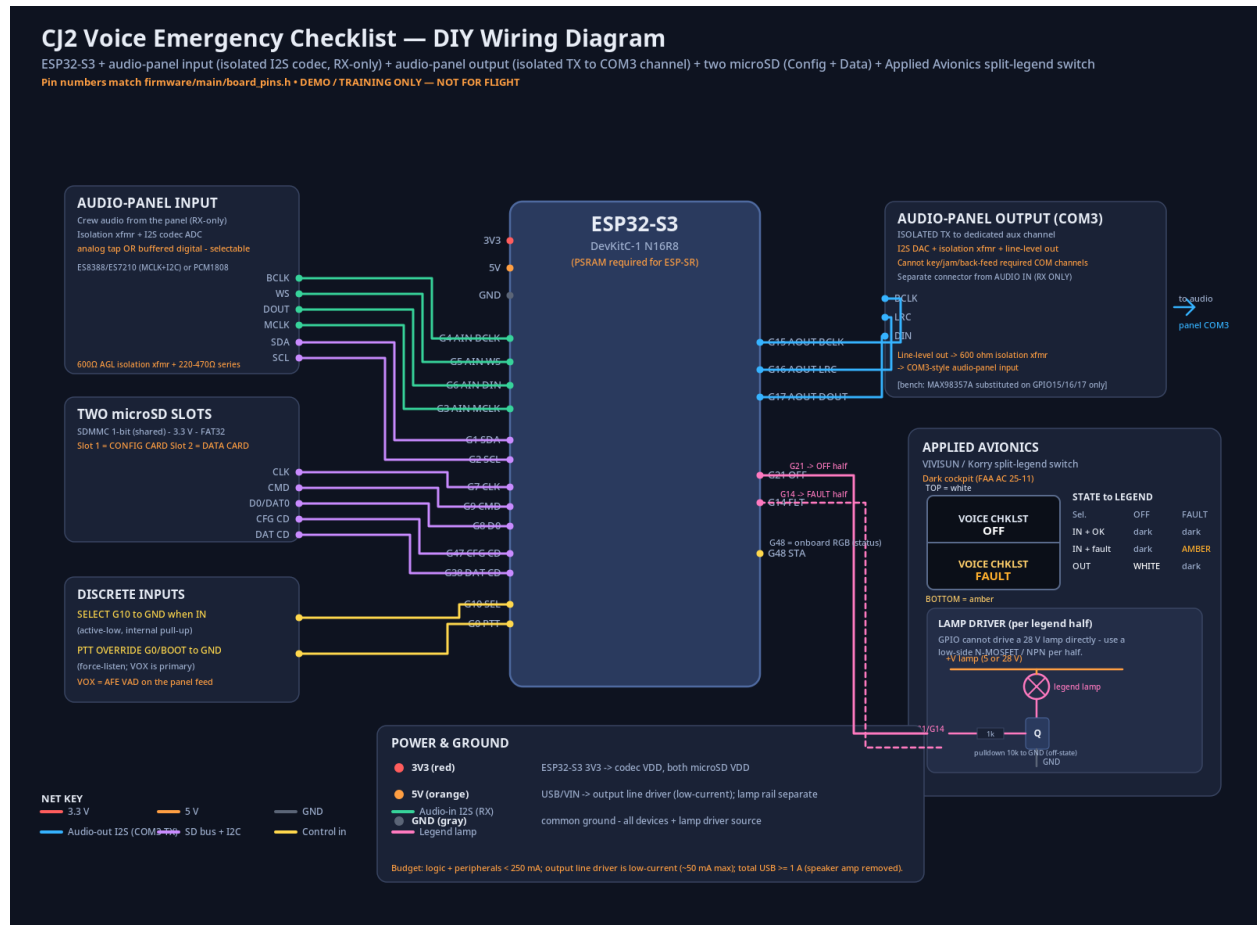
D.9 Bill of materials (DIY build)

Qty	Part	Spec / example
1	ESP32-S3 DevKit	DevKitC-1 N16R8 (PSRAM)
1	Audio-panel input codec	I2S codec ADC with line-in: ES8388 / ES7210 (need MCLK+I2C) or PCM1808 / CS5343 (self-clocking)
1	Audio isolation transformer	600 Ω:600 Ω aviation audio ground-loop isolator (Allen Avionics AGL series)
1	Input network	~220–470 Ω series resistor + RC anti-alias for the analog tap
1	Audio output isolation transformer	600 Ω:600 Ω line-level isolation transformer for the TX output stage (e.g. Allen Avionics AGL series or equivalent) — galvanic barrier on the COM3 output line
1	Audio output DAC / line driver	I2S DAC IC (e.g. PCM5102A class) or I2S-in line driver for the COM3 output stage; select based on output impedance and level requirements for the panel's COM3 input
(bench only)	I2S amp (bench test)	MAX98357A breakout — bench-test use only; not installed
(bench only)	Speaker (bench test)	4–8 Ω, ≥ 2 W — bench-test use only; not installed
2	microSD card + breakout	FAT32 — Config Card (slot 1) + Data Card (slot 2)
1	Annunciator switch	Applied Avionics VIVISUN/Korry split-legend (or 2 LEDs for bench)
2	Lamp driver	logic-level N-MOSFET (2N7002/AO3400) or NPN (2N2222)
4	Resistors	1 kΩ ×2 (gate), 10 kΩ ×2 (pulldown)
2–4	Pull-ups	10 kΩ on SD CMD/DATO; codec I2C pull-ups if needed
—	Caps	0.1 μF per device/rail, 100 μF bulk near output line driver (if applicable)
1	PTT button	momentary SPST (or use BOOT) — VOX override

Qty	Part	Spec / example
(opt.)	INMP441 MEMS mic	bench-test input only, not the installed source

Integrated alternative: ESP32-S3-Korvo-2 (~\$45–55) provides codec/SD on-board; use its BSP pin map and codec (ES8311/ES7210) init, repurpose its line-in for the audio-panel receive feed, and add the isolated COM3 output stage externally (DAC/line driver + output isolation transformer). A production unit adds the second card slot.

D.10 System wiring diagram



Complete system wiring — ESP32-S3, audio input, amp, two microSD cards (Config + Data), and split-legend annunciator with per-half lamp drivers.

PART E

Enclosure Specification

For the fabricating engineer. The housing is generic — sized for the electronics and the panel interface, not for a specific airframe. The only airframe-specific item is the mounting variant (DZUS slot vs. round instrument hole) and the card inside it.

E.1 Two-piece architecture

Piece	Contents	Where	Why
A. Panel bezel	Split-legend annunciator switch, PTT override button	Front panel / pedestal, on the DZUS rail	Crew must see/reach it; dark-cockpit annunciator in the normal scan. Speaker grille removed (no onboard speaker in the installed design).
B. Remote processor box	ESP32-S3, audio-input stage (isolation transformer + codec), audio output stage (DAC/line driver + isolation transformer for COM3 TX output), two microSD card slots, lamp-driver, power conditioning	Avionics bay / behind-panel, blind	Keeps heat, the card slots, and wiring out of the panel; close to the audio-panel tap and COM3 output point

A single all-in-one box is acceptable for a pure bench demo, but the two-piece split mirrors real remote-mount avionics and keeps the audio-input stage near the panel tap.

E.2 Piece A — panel bezel

- DZUS rail mount: fastener pitch 3/8 in (9.525 mm); clearance hole 0.255 in (6.48 mm); bezel height a whole multiple of 3/8 in (target 3-unit = 28.575 mm — the speaker cone space is freed so the 4-unit height is no longer needed unless a larger switch requires it; confirm against the chosen switch datasheet); standard pedestal width $\approx$ 146 mm aluminum (144.45 mm face); backplate 1/16 in (1.6 mm) 6061-T6; first/last fastener 14.29 mm from each end.
- Round-hole variant: fits a standard 3-1/8 in (79.4 mm) instrument cutout with four 6-32 screws on the standard bolt circle, to replace a blanking plate where no DZUS slot exists.
- Face layout (top→bottom): split-legend annunciator switch (cut per the *specific* switch datasheet — typical VIVISUN bezel $\approx$ 15×15 mm to 19×19 mm; top half VOICE CHKLST OFF white, bottom VOICE CHKLST FAULT amber, upright when installed); optional guarded/recessed PTT. No speaker grille — the onboard speaker is removed; checklist audio plays through the crew headsets via the COM3 output channel. The bezel height may be reduced from the prior 4-unit to 3-unit (28.575 mm) since the speaker cone space is freed.
- Material/finish: 6061-T6 aluminum 2.0–3.0 mm (or ABS/PC for a non-structural demo); matte black, low-gloss ($\leq$ 10 gloss units) to suppress glare; legend by the switch's internal engraving (preferred) or laser-etch + white/amber paint-fill; edges chamfered 0.5 mm.

E.3 Piece B — remote processor box

- Envelope: sized around the ESP32-S3 DevKitC-1 ($\approx 70 \times 26$ mm) plus the audio-input stage (codec + input isolation transformer), the audio output stage (DAC/line driver + output isolation transformer), two microSD breakouts, and the 2-channel lamp-driver; practical outer $\approx 120 \times 85 \times 45$ mm (unchanged — the output stage is compact; confirm against the actual stacked board set with both transformers). Note: without the speaker-amp the box is likely lighter and slightly cooler.
- Mounting: internal standoffs / M2.5 brass inserts — boards screwed down, not floating (vibration). Keep the audio-input and output stages and their shielded cabling away from the switching DC-DC; route input and output audio cables separately, both shielded; the isolation transformer mounts solidly (it is a magnetic part).
- Card slots: two externally-swappable microSD carriers, clearly and distinctly labeled CONFIG CARD — FAT32 (slot 1) and DATA CARD — FAT32 (slot 2), keyed or spaced so they cannot be confused/swapped; both swappable without opening the box. The Config Card carrier should accept a write-protect-locked card.
- Access & connectors: covered/recessed USB-C service port (bench use only); one keyed, positive-latching main connector (small MIL-circular or D-sub) carrying SELECT, both legend drives, PTT, power/ground (pinout from board_pins.h); plus a separate, shielded, clearly-labeled AUDIO IN — ISOLATED, RX ONLY connector for the audio-panel tap — kept on its own keyed connector so it can never be mis-mated to power, with the input isolation transformer inside the box on the panel side of the codec; and a separate, shielded, clearly-labeled AUDIO OUT — ISOLATED (COM3) connector for the isolated TX output to the audio-panel COM3-style channel — on its own keyed connector, physically distinct from the input connector, with the output isolation transformer inside the box on the panel side of the output stage. The two audio connectors must be clearly distinct and cannot be mis-mated. Accept USB 5 V $\geq$ 1 A; optional internal 28 V $\rightarrow$ 5 V DC-DC ($\geq$ 1 A; speaker-amp is removed) with TVS + fuse if a 28 V bus mock-up is wanted (mark as demo regulator, not DO-160 qualified).
- Material/EMI: aluminum preferred (doubles as EMI shield + heatsink); if plastic, add a grounded conductive shield liner/coating; single-point chassis ground stud bonded to the connector shell and ESP32 ground. Route the audio-in shield per the panel's grounding practice (often grounded only at the intercom — see C.3a).

E.4 Audio, thermal, environmental, labeling

- Audio: No onboard speaker in the installed design — speaker/grille removed from both pieces. Checklist read-aloud audio is delivered in-headset via the isolated COM3 output channel. If a bench-test speaker output is optionally fitted on the development unit (see D.3.3b), it may be wired to a header on the processor box only; no speaker cutout on the panel bezel.
- Thermal: ESP32-S3 + ESP-SR is low-power (a few hundred mW) — no fan. Passive convection (vent slots low/high) or conduction (thermal pad to the aluminum wall). If sealed, verify internal rise < 20 °C above 55 °C ambient. DC-DC (if fitted) on its own thermal path.
- Environmental: design toward the DO-160G categories in C.4 (Cat A2 temp, Cat S vibration, etc.) — design guidance for the prototype, formal test for a productized unit.
- Labeling: placard DEMO / TRAINING ONLY — NOT FOR FLIGHT; box exterior carries unit name, serial/asset field, the two card-slot labels (CONFIG CARD / DATA CARD, FAT32), the AUDIO IN — ISOLATED, RX ONLY connector marking (receive-only input tap), the AUDIO OUT — ISOLATED, COM3 connector marking (TX output to audio-panel COM3 channel), and the USB “bench use only” note; annunciator legends VOICE CHKLST OFF (white) / VOICE CHKLST FAULT (amber); amber = caution, white = status per AC 25-11B.

E.5 Deliverables & open items for the engineer

Deliverables: STEP + native 3D CAD of both pieces (boards + switch modeled in place); fully-dimensioned 2D drawings (DZUS pattern, switch cutout from the chosen datasheet, connector cutouts; GD&T on the switch cutout and DZUS holes); connector pinout mapped to board_pins.h; an FDM/SLA printable prototype for fit-check; a mechanical BOM; tolerances (switch cutout ± 0.1 mm, DZUS holes ± 0.1 mm on the 9.525 mm pitch, general ± 0.25 mm).

Open items (confirm before CAD): the exact Applied Avionics switch part number — the single most critical dimension; nothing finalizes until it is fixed. Also: DZUS slot vs. 3-1/8 in round hole in the target panel; the audio-panel tap point, level, and grounding for the target installation (sets the input isolation-transformer + divider design); the codec part for the digital path (sets the I2C/MCLK init); the audio-panel COM3-style output channel injection point, impedance, and level requirements (sets the output isolation transformer and line driver design — this is a critical open item for the output stage); the output DAC/line driver part (sets the COM3 output stage component selection); whether a 28 V input is wanted. No speaker — the speaker is removed.

Reference dimensions: DZUS pitch 9.525 mm · DZUS hole 6.48 mm · backplate 1.6 mm · first fastener offset 14.29 mm · pedestal panel width ≈ 146 mm · round instrument hole 79.4 mm · remote box $\approx 120 \times 85 \times 45$ mm · two microSD slots · isolated AUDIO IN (RX ONLY) connector (input tap) · isolated AUDIO OUT (COM3) connector (TX output to audio-panel COM3 channel) · no onboard speaker in the installed design.

PART F

References

All cited regulatory, component, and mechanical sources, with clickable links.

Certification & regulatory

#	Reference	Link
1	FAA Policy Statement PS-AIR-21.8-1602 — “Approval of Non-Required Safety Enhancing Equipment (NORSEE)”	open
2	FAA Technical Standard Orders (TSO) program overview	open
3	FAA AC 21-16G — acceptance of RTCA/DO-160 versions D–G	open
4	RTCA DO-160G — Environmental Conditions and Test Procedures for Airborne Equipment	open
5	FAA AC 20-152A — Development Assurance for Airborne Electronic Hardware (simple-hardware relief)	open
6	FAA AC 20-168 / RTCA DO-313 — installation of non-essential, non-required equipment	open
7	TSO / TC / STC / PMA primer (Afuzion)	open
8	FAA AC 25-11B — electronic flight displays; annunciation color / dark-cockpit conventions	open

Hardware & components

#	Reference	Link
1	Allen Avionics AGL audio ground-loop isolation transformers (analog audio-panel tap)	open
2	Everest-Semi ES8388 audio codec (line-in I2S ADC, MCLK + I2C)	open
3	ES7210 multichannel audio ADC (Espressif-supported codec)	open
4	TI PCM1808 stereo audio ADC (self-clocking I2S input)	open
5	Espressif ESP-SR AFE / VAD (voice-activity detection for VOX)	open
6	INMP441 microphone datasheet (bench-test option only)	open
7	MAX98357A amplifier datasheet (Analog Devices) (bench-test amp only)	open
8	MAX98357A breakout guide (Adafruit) (bench-test amp only)	open
9	ESP32-S3 GPIO drive current discussion	open
10	microSD operating current	open
11	Applied Avionics VIVISUN lighted pushbutton switches	open
12	Espressif ESP-SR speech framework	open
13	Espressif ESP32-P4 product page	open
14	ESP32-P4 vs ESP32-S3 performance	open

Mechanical / panel

#	Reference	Link
1	DZUS panel-building dimensions guide (MyCockpit)	open
2	Standard 3-1/8 in instrument cutout (Aircraft Spruce)	open

DEMO / TRAINING ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS. Repository: github.com/flas-tech/cj2-voice-emergency-checklist (MIT License). The certification sections describe a path, not held approvals; no NORSEE LOA, STC, PMA, or TSOA exists for this article.